

SIGNEX — SIGNAL EXTRACTION & ANALYSIS ENGINE

NTRO ID26147 CLASSIFIED INTELLIGENCE DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: 05b042c4a365f45a

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : no84.2021-12-15T22_38_34_739.wav
File SHA-256 Digest : 05b042c4a365f45a883e56a4410122ed099e1cea99a6ba900a0dc74578486c21
Payload File Size : 4,895,352 bytes (4780.6 KB)
Analysis Timestamp : 2026-09-19T08:45:35Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	24000.0 Hz	Baseband Nyquist Rate
Center Frequency (F_c)	+0.518 kHz	Spectral Centroid Offset
Estimated Baud Rate	1.51 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	17.6 Hz	Half-Power Frequency Span
Occupied Bandwidth (99%)	23.51 kHz	Cumulative PSD 99% Energy
Estimated SNR	10.99 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	19.00 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	-455.57 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.4453	Wiener Entropy Index (0.0 – 1.0)

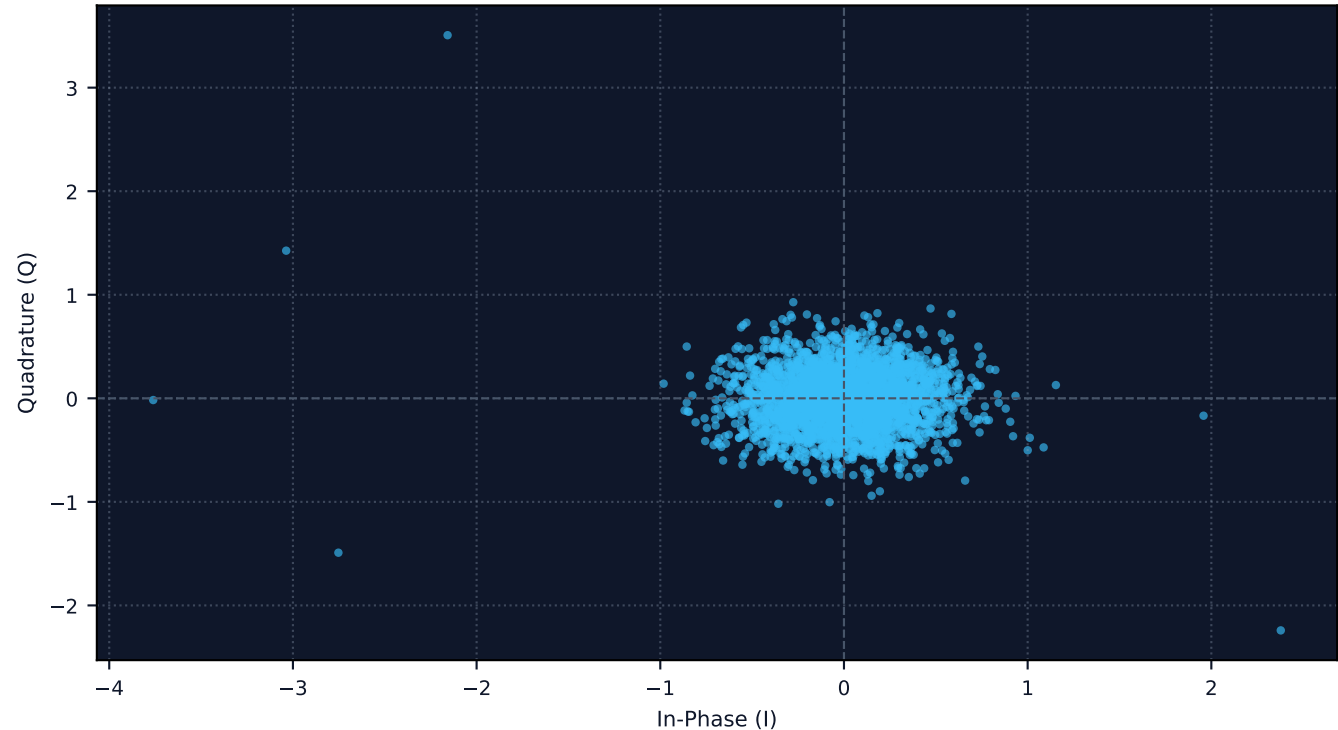
PREDICTED MODULATION : AM-DSB

CLASSIFIER CONFIDENCE : 82.63% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: BPSK | Recovered Bits: 152,973 | Estimated EVM: 9.97% | CRC Validated Frames: 0

Demodulated Baseband Constellation Diagram — [AM-DSB]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	82	0x0000	0	19104 B	CRC MISMATCH	PARTIAL_RECOVER
#2	336	0x0000	1	19072 B	CRC MISMATCH	PARTIAL_RECOVER
#3	348	0x0000	2	19071 B	CRC MISMATCH	PARTIAL_RECOVER
#4	1525	0x0000	3	18924 B	CRC MISMATCH	PARTIAL_RECOVER
#5	2032	0x0000	4	13363 B	CRC MISMATCH	CRC_MISMATCH

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 A0C908B1F14F5F56D0C1B0AAAE043CE2 [FRAME 1 PAYLOAD]